# AQ3.2: Activity Questions 2 - Not Graded

This assignment will not be graded and is only for practice.
**Level 1**

If $(1, 2, 3) + v = (3, 2, 1)$ in the vector space $R^3$ with usual addition and scalar multiplication, where $v = (a, b, c)$, then find the value of $a + b + c$.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
(Type: Numeric) 0
*1 point*
Which of the following statements are related to the cancellation law for vectors $x, y, z$ in a vector space.

Statement P: $x + y = z + y \implies x = z$.
Statement Q: $x + y = z + y \implies x = y$.
Statement R: $x + y = z + y \implies y = z$.
Statement S: $x + y + z = x \implies y + z = 0$

Find the number of correct statements

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
(Type: Numeric) 2
*1 point*
*1 point*
Which of the following options is/are true for a non zero vector space $V$ ?

- [ ] $x + y = z + y \implies x = z \ \forall x, y, z \in V$.
- [ ] $ax = bx, \ \forall \ x \in V \implies a = b,$ where $a, b \in R$.
- [ ] $ax = ay, \ \forall \ a \in R \implies x = y,$ where $x, y \in V$.
- [ ] None of the above.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
$x + y = z + y \implies x = z \ \forall x, y, z \in V$.
$ax = bx, \ \forall \ x \in V \implies a = b,$ where $a, b \in R$.
$ax = ay, \ \forall \ a \in R \implies x = y,$ where $x, y \in V$.
*1 point*
Which of the following options is/are true for a non zero vector space $V$ ?

- [ ] $av = 0, \ \forall \ v \in V \implies a = 0$.
- [ ]

$av = 0, \forall\, a \in R \implies v = 0$av = 0, ∀ a ∈ R ⟹ v = 0.

☐

$c \in R$c ∈ R and $v, w \in V$v, w ∈ V such that, $c(v - w) \neq cv - cw$c(v − w) = cv − cw.

☐

$c((v + w) - (v - w) - 2w) = 0\ \forall c \in R,$ and $v, w \in V$c((v + w) − (v − w) − 2w) = 0 ∀c ∈ R, and v, w ∈ V

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$av = 0, \forall\, v \in V \implies a = 0$av = 0, ∀ v ∈ V ⟹ a = 0.

$av = 0, \forall\, a \in R \implies v = 0$av = 0, ∀ a ∈ R ⟹ v = 0.

$c((v + w) - (v - w) - 2w) = 0\ \forall c \in R,$ and $v, w \in V$c((v + w) − (v − w) − 2w) = 0 ∀c ∈ R, and v, w ∈ V

Let $(3, 2, -3) \in R^3$(3, 2, −3) ∈ R3 be a vector from the vector space $R^3$R3 with usual addition and scalar multiplication. If $c(3, 2, -3) = (-6, -4, 6)$c(3, 2, −3) = (−6, −4, 6), find the value of $c$c.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) -2

*1 point*

*1 point*

Which of the following options is/are true?

☐ The zero vector is a unique vector in a vector space.

☐ Inverse of a vector is a unique vector in a vector space.

☐

$0v \neq 00$v = 0, where $v$v is a vector from a vector space of $R$R.

☐

$c0 = 0$c0 = 0, where $c \in R$c ∈ R and 0 is the zero vector from a vector space over $R$R.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

The zero vector is a unique vector in a vector space.

Inverse of a vector is a unique vector in a vector space.

$c0 = 0$c0 = 0, where $c \in R$c ∈ R and 0 is the zero vector from a vector space over $R$R.

**Level 2**

Let $V$V be a plane plane parallel to the $XY$XY -plane. Any plane parallel to $XY$XY -plane is given by $z = c$z = c. We define addition of two vectors $v_1 = (x_1, y_1, c)$v1 = (x1, y1, c) and $v_2 = (x_2, y_2, c)$v2 = (x2, y2, c) on $V$V as follows: First project $v_1$v1 and $v_2$v2 on the $XY$XY -plane (we will get the vectors $(x_1, y_1, 0)$(x1, y1, 0) and $(x_2, y_2, 0)$(x2, y2, 0) by projection on the $XY$XY -plane) and then calculate the addition of the vectors we obtained by the projection on the $XY$XY -plane (we will obtain $(x_1 + x_2, y_1 + y_2, 0)$(x1 + x2, y1 + y2, 0)). Then project the obtained vector back to the plane $V$V (we will obtain the vector $(x_1 + x_2, y_1 + y_2, c)$(x1 + x2, y1 + y2, c)).

Let $V$V be a plane which is parallel to the $XY$XY -plane, defined by $z = 2$z = 2. Let $v_1 = (1, 2, 2)$v1

$= (1, 2, 2)$ and $v_2 = (0, 3, 2)$ be in $V$. Answer questions 7 and 8.

If $v_1^{'} = (a_1, b_1, c_1)$ and $v_2^{'} = (a_2, b_2, c_2)$ are the projections of $v_1$ and $v_2$ on the $XY$ - plane, respectively, then find the value of
$2(a_1 + b_1 + c_1) - (a_2 + b_2 + c_2)$.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
(Type: Numeric) 3
*1 point*
If $v_1 + v_2 = (a_1, b_1, c_1)$ as per the addition defined above, then find the value of $(2a_1 + b_1 - 2c_1)$.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
(Type: Numeric) 3
*1 point*

Let $V$ be a plane plane parallel to the $XY$ -plane. Any plane parallel to the $XY$ -plane is given by $z = c$. We define scalar multiplication of a vector $v = (x, y, c)$ on $V$ as follows: First project $v$ on the $XY$ -plane first (we will get the vector $(x, y, 0)$ by projection on the $XY$ -plane) and then calculate the scalar $(\alpha)$ multiple of the vector we obtained by the projection on the $XY$ -plane (we will obtain $(\alpha x, \alpha y, 0)$). Then project the obtained vector back to the plane $V$ (we will obtain the vector $(\alpha x, \alpha y, c)$).
Let $V$ be a plane which is parallel to the $XY$ -plane, defined by $z = 2$. Let $v = (1, 2, 2)$ be in $V$. Answer questions 9 and 10.

If $v^{'} = (a_1, b_1, c_1)$ is the projection of $v$ on the $XY$ - plane, then find the value of $(a_1 + 2b_1 + c_1)$.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
(Type: Numeric) 5
*1 point*
If $4v = (a_1, b_1, c_1)$ as per the scalar multiplication defined above, then find the value of $(a_1 + 2b_1 + c_1)$.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
(Type: Numeric) 22
*1 point*

Check Answers
Your score is: 0/10